1 Verification

Each v0.1 report in the table can be checked with one command, from an empty directory,

```
uvx agent-evidence-vectors==0.32.1 -read-w3c-report <file>
```

which prints `conforms`, or one line per shape error or rejection row, and exits 0 when the report conforms, 1 when it does not, and 2 when the reader reached no verdict. Rows that are not reports (readers' verdicts, comparisons, census counts) are checked by their rerun command instead. The evidence classes are their authors' words: "reproduction in O8's sense" (Oct/0163), "a reader written from the 0087 text alone" and "proposed" (Oct/0162), "the closest this paper comes to an independent record" (Oct/0150). The class of every cell this appendix runs is "automated rerun from the pins, operated by the appendix's author". Following Oct/0163, units differ by row and no column is summed.

Table 1: The verification column. Commands run in the appendix directory unless they start with `uvx`.

Row	Command	Verdict	Unit	Denominator	Evidence class
R1	`-read-w3c-report report.json`	conforms	checks	272	reproduction in O8's sense (Oct/0170, Oct/0171)
R2	`-read-w3c-report` on each of the three reports	conforms	checks	72, 168, 750	reproduction in O8's sense (Oct/0064)
R3-v01	`-read-w3c-report run-report-v01.json`	conforms	checks	48	producer's converter (Nov/0000)
R4	`-read-w3c-report` on each of the three reports	conforms from v0.18.1; rejected under W3C-R-020 before v0.18	checks	76, 67, 16	the closest to an independent record, through the producer's glue (Oct/0150)
R4-driver	`python rerun.py R4-driver`	equal to the recorded run	cases	as recorded	without the glue (Nov/0003)
R5	`-read-w3c-report finality-threshold-report.level1.json`	conforms	checks (six thresholds per generated draw, plus six exclusions)	150,006	reproduction in O8's sense (Oct/0083, Oct/0084)
R6	`python rerun.py R6`	reproduced; no report file	tests	4 baseline, 6 claim-boundary	reproduction in O8's sense (Oct/0018)
R6, re-encoded	`-read-w3c-report run-report-v01.json`	conforms	checks	4	producer's converter (Nov/0006)
R7	`python rerun.py R7`	reproduced; a native packet, not a v0.1 report	outcomes	21	reproduction in O8's sense (Oct/0085, Oct/0184)
R8	`python rerun.py R8`	reproduced	members	169 judged	a reader written from the 0087 text alone
R8-compare	`python rerun.py R8-compare`	92 of 96; 23, 2, 44 of 69	members	96 and 69	a comparison (Oct/0188, Nov/0002)
R9	`python rerun.py R9`	83/82 at v0.17.4; 41/124 at v0.18.0 and v0.18.1	members	165	a comparison across releases
R10	`python rerun.py R10` (added)	bytes as posted	values	1	posted measurement (Oct/0038, Oct/0046)
R11	`python rerun.py R11`	output digest as posted	members	165	a reader written from the text
R12	`python rerun.py R12`	per-member verdicts as posted	members x paths	33 x 3	proposed
R13	`python rerun.py R13`	every listed file OK	synthetic cases	5	reproduction in O8's sense (Oct/0179); proposed
R14	`python matrix/matrix.py run`	see the matrix below	verdicts	reports x readers	a comparison
R15	none posted	tally only: 3 of 592	artifacts	592	the producer's own count (Oct/0162)

Every report under every reader. The committed reader matrix retains the original R6 report and the separately re-encoded R6-v01 report (Nov/0006) as different inputs. The generated table below lists the report

Reader	R1	R2-v0.2	R2-v0.3	R2-v0.4	R3-v01	R4-heldfast	R4-native	R4-map	R5	R6	R6-v01
Python 0.12.0	C	C	C	C	C	R R-020	R R-020	R R-020	C	R shape	C
Python 0.13.0	C	C	C	C	C	R R-020	R R-020	R R-020	C	R shape	C
Python 0.17.4	C	C	C	C	C	R R-020	R R-020	R R-020	C	R shape	C
Python 0.18.1	C	C	C	C	C	C	C	C	C	R shape	C
Python 0.31.0	C	C	C	C	C	C	C	C	C	R shape	C
Go v0.18.1	C	C	C	C	C	C	C	C	C	R shape	C
Go main	C	C	C	C	C	C	C	C	C	R shape	C
Rust v0.18.1	C	C	C	C	C	C	C	C	C	R shape	C
Rust main	C	C	C	C	C	C	C	C	C	R shape	C
0039 reader	C	C	C	C	C	C	C	C	C	N	C
Kroehl 6d8ee54b	C	C	C	C	C	C	C	C	C	T	C

Table 2: Each report rerun from its pins, read by each reader. C conforms; R rejected, with the rows cited (a numbered row of the rejection table by its number, an unnumbered rule as R- and its identifier's number); T cannot tell; N cannot read. R7 has no column: its packet is not a v0.1 report.

Reader	Mutants	Rejected	Conforms	Cannot tell	Cannot read	Target cited
Python 0.12.0	168	161	7	0	0	160/168
Python 0.13.0	168	161	7	0	0	161/168
Python 0.17.4	168	161	7	0	0	161/168
Python 0.18.1	168	168	0	0	0	168/168
Python 0.31.0	168	168	0	0	0	168/168
Go v0.18.1	168	168	0	0	0	168/168
Go main	168	168	0	0	0	168/168
Rust v0.18.1	168	168	0	0	0	159/168
Rust main	168	168	0	0	0	159/168
0039 reader	168	61	88	18	1	58/59
Kroehl 6d8ee54b	168	82	37	49	0	58/59

Table 3: Single-field negative controls: every reader on every mutant. Target cited counts mutants whose target row the reader named, over those whose target its labels can name.

and reader inventory, their stored verdicts and the per-reader negative-control totals directly from `matrix.json`. R6-v01 conforms in every stored reader cell; that does not change the original R6 outcomes: Python, Go and Rust reject, Rouiada Abbas's reader cannot read, and Lars Kersten Kroehl's reader cannot tell. The Python CLI result at 0.32.1 reported in Nov/0010 is a separate observation, not another reader column in this matrix. R4's reports are rejected under W3C-R-020 by the older Python readers. The readers written from the text receive each report wrapped as a corpus member, through an adapter that writes `{"id": <input>, "subject": <report>}` and nothing else. R14 is a comparison of reader verdicts, not an independent reproduction of producer checks.

Reader availability in R9. The release comparison uses Python and Go at v0.17.4; Rust is available from v0.18.0. It does not establish a Rust result at v0.17.4 (Nov/0010).

- R1: `uvx agent-evidence-vectors==0.32.1 -read-w3c-report report.json`
- R2-v0.2: `uvx agent-evidence-vectors==0.32.1 -read-w3c-report run-residue-v0.2.json`
- R2-v0.3: `uvx agent-evidence-vectors==0.32.1 -read-w3c-report run-residue-v0.3.json`
- R2-v0.4: `uvx agent-evidence-vectors==0.32.1 -read-w3c-report run-residue-v0.4.json`
- R3-v01: `uvx agent-evidence-vectors==0.32.1 -read-w3c-report run-report-v01.json`
- R4-heldfast: `uvx agent-evidence-vectors==0.32.1 -read-w3c-report heldfast-profiles.json`
- R4-native: `uvx agent-evidence-vectors==0.32.1 -read-w3c-report native-verify-mjs.json`
- R4-map: `uvx agent-evidence-vectors==0.32.1 -read-w3c-report probity-map-reader.json`
- R5: `uvx agent-evidence-vectors==0.32.1 -read-w3c-report finality-threshold-report.level1.json`
- R6: `uvx agent-evidence-vectors==0.32.1 -read-w3c-report records.v0.1.json`
- R6-v01: `uvx agent-evidence-vectors==0.32.1 -read-w3c-report run-report-v01.json`

Table 4: Retained controls by target population. Outcomes are rejected / conforms / cannot tell / cannot read / not run. Positive controls are separate from negative-control totals.

Reader	Population	Inputs	Outcomes
Python 0.12.0	Numbered rows	59	59/0/0/0/0
Python 0.12.0	Unnumbered rules	90	83/7/0/0/0
Python 0.12.0	Shape	19	19/0/0/0/0
Python 0.12.0	Positive controls	7	7/0/0/0/0
Python 0.13.0	Numbered rows	59	59/0/0/0/0
Python 0.13.0	Unnumbered rules	90	83/7/0/0/0
Python 0.13.0	Shape	19	19/0/0/0/0
Python 0.13.0	Positive controls	7	7/0/0/0/0
Python 0.17.4	Numbered rows	59	59/0/0/0/0
Python 0.17.4	Unnumbered rules	90	83/7/0/0/0
Python 0.17.4	Shape	19	19/0/0/0/0
Python 0.17.4	Positive controls	7	7/0/0/0/0
Python 0.18.1	Numbered rows	59	59/0/0/0/0
Python 0.18.1	Unnumbered rules	90	90/0/0/0/0
Python 0.18.1	Shape	19	19/0/0/0/0
Python 0.18.1	Positive controls	7	0/7/0/0/0
Python 0.31.0	Numbered rows	59	59/0/0/0/0
Python 0.31.0	Unnumbered rules	90	90/0/0/0/0
Python 0.31.0	Shape	19	19/0/0/0/0
Python 0.31.0	Positive controls	7	0/7/0/0/0
Go v0.18.1	Numbered rows	59	59/0/0/0/0
Go v0.18.1	Unnumbered rules	90	90/0/0/0/0
Go v0.18.1	Shape	19	19/0/0/0/0
Go v0.18.1	Positive controls	7	0/7/0/0/0
Go main	Numbered rows	59	59/0/0/0/0
Go main	Unnumbered rules	90	90/0/0/0/0
Go main	Shape	19	19/0/0/0/0
Go main	Positive controls	7	0/7/0/0/0
Rust v0.18.1	Numbered rows	59	59/0/0/0/0
Rust v0.18.1	Unnumbered rules	90	90/0/0/0/0
Rust v0.18.1	Shape	19	19/0/0/0/0
Rust v0.18.1	Positive controls	7	0/7/0/0/0
Rust main	Numbered rows	59	59/0/0/0/0
Rust main	Unnumbered rules	90	90/0/0/0/0
Rust main	Shape	19	19/0/0/0/0
Rust main	Positive controls	7	0/7/0/0/0
0039 reader	Numbered rows	59	58/1/0/0/0
0039 reader	Unnumbered rules	90	3/78/9/0/0
0039 reader	Shape	19	0/9/9/1/0
0039 reader	Positive controls	7	0/7/0/0/0
Kroehl 6d8ee54b	Numbered rows	59	58/0/1/0/0
Kroehl 6d8ee54b	Unnumbered rules	90	20/28/42/0/0
Kroehl 6d8ee54b	Shape	19	4/9/6/0/0
Kroehl 6d8ee54b	Positive controls	7	0/7/0/0/0

Negative controls for the readers. A reader that accepted everything could match the conforming base reports, so the matrix also records single-field negative controls. The generated controls table gives the denominator, outcomes and target-citation counts for each reader. Positive controls and targets marked not applicable are separate populations; neither belongs in that negative-control denominator. The positive controls use a non-ASCII check identity and a root recomputed over the RFC 8785 form. Their ASCII-escaped twins target W3C-R-020, the cause Kenne Ives gave for R4 (Oct/0150). A declared moved that disagrees with the recomputed value targets

W3C-R-021; a recomputed moved outside compared targets W3C-R-016.

The readers written from the text were built to answer cannot-tell where the text leaves the reading open (Oct/0039), and the controls show where that stops: over all negative controls each also accepts copies aimed at requirements outside numbered rows 1 to 14, and a reader that computes no tree root cannot reject a changed root (Nov/0028). The generated target-population table retains the separate outcome classes, including positive controls. The older Python readers accept the ASCII-escaped W3C-R-020 twins and reject their RFC 8785 positive controls. For the text-derived readers, the shared numbered-row control not rejected is R1 with a qualifier on a non-verdict check, targeting row 6. Detailed cells are retained in the generated `matrix/matrix.md`. The command list gives ways to check the reports; it is not an additional set of recorded reader cells. Target cited counts whether the named target was cited, not merely whether a mutant was rejected. These are findings for their authors, not faults.